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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Made in UTM
Project / Solution Name	Laluan: a career GPS for the AI era
Name (Leader)	Tan Hong Sheng
Email & Phone (Leader)	tanhongsheng050204@gmail.com (phone as on the registration form)
University Affiliation(s)	Universiti Teknologi Malaysia (UTM)
Live Prototype URL	https://laluan-psi.vercel.app (API: https://laluan-api.onrender.com/docs)
Code Repository URL (public / access granted to judges)	https://github.com/tanhongsheng050204-code/SDC-HACKATHON-2026---MADE-IN-UTM (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The user opens a React single-page app on **Vercel**. It calls a **FastAPI** backend over HTTPS, running as a **Docker** container on **Render** (Singapore). Profiles, progress, trend snapshots and optional logins live in **Supabase** (Postgres and Auth, Singapore).

A CV is read in memory. Scans go through our own OCR (on **RunPod Serverless**, else on the server), then names, emails, phone and IC numbers are removed. An open model, **Qwen2.5-7B-Instruct on RunPod**, reads skills from the scrubbed text, and **faster-whisper on RunPod** transcribes voice input. **Google Gemini** writes coach answers, interview feedback and summaries; every sentence is checked against our sources before it is shown. Without keys, rules and templates take over.

Matching, forecasts, scenarios and pathways are plain Python over versioned **JSON reference data** (WEF, O*NET, courses, SDG map), with no AI call.

GitHub Actions runs the tests on every push and a weekly trend refresh that collects arXiv/OpenAlex, freehire.me job ads, Hacker News, Stack Overflow, GitHub, npm, PyPI and OpenDOSM. Each new dated snapshot goes live only after an admin approves it. A diagram is in the appendix.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript, Python 3.12, CSS, SQL (Postgres; SQLite locally).	Programming languages	—
React 18.3.1, React DOM 18.3.1	User interface	MIT
TypeScript 5.6.3	Typed frontend code	Apache-2.0

Name & Version	Purpose in Project	License
Vite 5.4.11, @vitejs/plugin-react 4.3.4	Frontend dev server and build	MIT
@supabase/supabase-js 2.45.4	Google or Microsoft login in the browser	MIT
qrcode 1.5.4	QR code on the shareable Future Card	MIT
@fontsource-variable/inter 5.3.0 (Inter font)	Self-hosted typeface	SIL OFL 1.1
FastAPI 0.148.0 (Starlette 1.9.0)	HTTP API	MIT / BSD-3
Pydantic 2.14.1	Request validation (also rejects demographic fields)	MIT
Uvicorn 0.54.0	ASGI server	BSD-3
psycopg 3.3.6	Supabase Postgres connection	LGPL-3.0
PyJWT 2.15.1	Checking Supabase login tokens	MIT
google-genai 1.75.0	Gemini API client	Apache-2.0
HTTPX 0.28.1	Calls to RunPod and the trend sources	BSD-3
pypdf 6.21.1	Reading uploaded PDF CVs in memory	BSD-3
RapidOCR 2.1.0, ONNX Runtime 1.31.0, pypdfium2 5.14.0, Pillow 13.0.0	Our own OCR for scanned CVs and photos (RunPod worker first, else the API server)	Apache-2.0 / MIT / Apache-2.0 / MIT-CMU
runpod SDK 1.7	The OCR worker's serverless handler	MIT
python-multipart 0.0.32	CV file uploads	Apache-2.0
sentry-sdk 2.72.0, @sentry/react 8.38.0	Optional error reports, personal data removed	MIT

Name & Version	Purpose in Project	License
pytest 8.4.2, Vitest 2.1.8, Testing Library, axe-core 4.10.2	Tests and accessibility checks (development only)	MIT / MIT / MIT / MPL-2.0
PyMuPDF	One-off script that reads WEF Figure 2.3 from the report PDF; not part of the app	AGPL-3.0
Python 3.12-slim, Node 20, nginx 1.27 (Docker images)	API container and local docker compose run	PSF / MIT / BSD-2

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Gemini API (Google): gemini-3.8-flash, gemini-embedding-001	Coach answers, interview feedback, proof-check questions, summaries, embeddings for unfamiliar skill phrases. The app runs without it.	Free
RunPod Serverless API	Qwen2.5-7B-Instruct reads skills from scrubbed CV text; faster-whisper transcribes voice; our OCR worker reads scans	Sponsor-provided credits
Supabase (Postgres and Auth)	Database and optional Google or Microsoft login	Free
arXiv API (Cornell University)	AI research papers per capability, for trend signals	Free
OpenAlex API (OurResearch)	Fallback paper counts, and AI papers with a Malaysian institution	Free
freehire.me public API	Malaysian job ads: skill and role counts, and the newest openings with links to the original postings	Free, no key
Hacker News Search API (Algolia)	Skill mentions in “Who is hiring?” threads	Free
Stack Exchange, GitHub, npm and PyPI Stats APIs	Developer attention per skill	Free
OpenDOSM / data.gov.my API (Department of Statistics Malaysia)	Labour force, employment by sector, consumer prices and salaries	Free
Gmail SMTP (Google)	Weekly digest and job-alert emails the user asked for	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel	Hosts the built React frontend (https://laluan-psi.vercel.app)	Hobby (free), global edge
Render	Runs the FastAPI backend as a Docker container (render.yaml, backend/Dockerfile)	Free web service, Singapore
Supabase	Postgres for profiles, progress, Future Cards, trend snapshots and review decisions (row-level security on); login through Supabase Auth	Free, Singapore
RunPod Serverless	Open models for CV reading (Qwen2.5-7B-Instruct on vLLM), voice (faster-whisper) and our OCR worker (runpod/ocr), so raw CV text and audio stay away from Google	Sponsor credits, pay per second
Replit	Development workspace and an alternative host for the same code	Sponsor-provided
GitHub and GitHub Actions	Public repository; tests on every push; scheduled jobs (keep-awake ping, weekly trend refresh, digest and job alerts)	Free
Docker / Docker Compose	One-command local run of the whole app (docker compose up --build)	Local

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 3.8 Flash (gemini-3.8-flash)	Google, via the Gemini API	Pre-trained, called by API	Gemini API Terms of Service
Gemini Embedding (gemini-embedding-001)	Google, via the Gemini API	Pre-trained, called by API	Gemini API Terms of Service
Qwen2.5-7B-Instruct	Alibaba Qwen team, served with vLLM on RunPod Serverless	Pre-trained, self-hosted	Apache-2.0
Whisper small (faster-whisper)	OpenAI / SYSTRAN, on RunPod Serverless	Pre-trained, self-hosted	MIT
PaddleOCR PP-OCRv4 English recogniser and RapidOCR text detector	PaddlePaddle / RapidAI, run in our OCR worker	Pre-trained, self-hosted	Apache-2.0

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
WEF Future of Jobs Report 2025, Figure 2.3 (net job growth 2025–2030, 126 occupations)	https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf	Public	WEF terms of use; cited on every role	N
O*NET 30.0 task statements	https://www.onetcenter.org/	Public	CC BY 4.0	N
UN SDG goals, targets and colours	https://sdgs.un.org/goals	Public	UN SDG guidelines	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM labour force, employment by sector, consumer prices, salaries; DOSM Salaries & Wages 2025	https://open.dosm.gov.my	Public	CC BY 4.0	N
arXiv and OpenAlex paper counts	https://export.arxiv.org/api/, https://openalex.org/	Public (counts only)	arXiv API terms; CC0	N
Malaysian job postings	https://freehire.me/docs/api	Public (counts stored; postings shown live with links)	freehire terms; project MIT	N
Hacker News “Who is hiring?” skill mentions	https://hn.algolia.com/api	Public (counts only)	HN / Algolia API terms	N
Stack Overflow, GitHub, npm and PyPI activity	Each service's public API	Public (counts only)	Each API's terms	N
Anthropic Economic Index	https://huggingface.co/datasets/Anthropic/EconomicIndex	Public (aggregates)	CC BY	N
OpenAI “GPTs are GPTs” occupation exposure scores	https://github.com/openai/GPTs-are-GPTs	Public	MIT	N
ILO Working Paper 140, Annex Table A1	https://www.ilo.org/publications/generative-ai-and-jobs-refined-global-index-occupational-exposure	Public	CC BY 4.0	N
MyCOL 2024/2025 (TalentCorp critical occupations)	https://www.talentcorp.com.my/mycol/	Public report	Cited with link	N
Course catalogue (prices, hours, HRD Corp status)	Providers' own pages, dated in backend/data/courses.json	Self-collected	Facts with links to the providers	N
Roles, skills and SDG map	backend/data/	Self-created	Apache-2.0 (this repository)	N
Demo personas, evaluation CVs, sample company (190 people) and class (52 students)	backend/data/	Synthetic	Apache-2.0 (this repository)	N (all invented)
Landing-page photos (3)	Unsplash: Gema Saputera, Mimi Thian, Vitaly Gariev	Public	Unsplash License	N (not linked to any data)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), with Claude Opus models	Planning documents; writing and refactoring backend and frontend code and tests; screen layouts (modelled on a reference site the team chose); Docker and deployment files; extracting WEF figures; BM and Chinese interface text; the demo GIFs, pitch deck and these declarations	Major
Google Gemini (inside the product)	Drafting automate / assist / human labels for O*NET tasks, each reviewed by a person	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team chose the problem framing, users, features, design reference and data sources, and directed every change. AI-written code ships only with tests: about 490 automated backend tests, frontend component tests, a type-checked build and lint, and CI on every push (on SQLite and Postgres). Each feature was clicked through in a browser. Each problem found, and the fix, is logged in REVIEW_LOG.md, and design choices with their alternatives are in DECISIONS.md. Inside the app, AI never acts alone: the coach only suggests, trend updates need an admin's approval, and AI-written text is checked against our sources.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
RunPod	Serverless GPU credits and API	Qwen2.5-7B-Instruct reads skills from scrubbed CV text, faster-whisper transcribes voice input, and our OCR worker (runpod/ocr) reads scanned CVs
Replit	Platform	Development workspace and an alternative host for the same code

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replaceable, by design. RunPod lets CV text and voice stay away from Google, with open models we choose, but if an endpoint is down or not set, Gemini or the dictionary reader reads the CV, the server's own OCR reads scans, and the browser's speech input replaces Whisper. Replit is one place the app runs; the same code runs on Vercel and Render, or locally with Docker. The core route (matching, forecasts, scenarios and pathways) is plain calculation and needs no sponsor resource.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Collected: confirmed skills, current role, years of experience, goals, SDG choices and course progress, under a random profile ID in Supabase (Singapore). An account (Google or Microsoft login) is optional; an email address is kept only while the user has emails or job alerts switched on. The API rejects age, gender and background fields.

CVs: read in memory and never stored. Names, emails, phone and IC numbers are removed before any AI (RunPod or Gemini) sees the text, and the user sees exactly what will be sent.

Retention: profiles not saved to an account are deleted after 30 days unused, demo copies after a day. Users can download or delete everything at any time.

Protection: HTTPS, row-level security, saved profiles open only through their login, logged admin actions, no IP addresses or profile IDs in logs. HR uploads with personal columns are refused; results are totals only.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

- **Hallucination:** AI text could invent courses or numbers. Mitigation: courses, roles and figures come only from reference data, and every AI sentence is checked against those sources and rewritten or removed if unsupported.
- **Forecast uncertainty:** WEF figures are global employer-survey estimates. Mitigation: Malaysian signals, three scenarios, 80% ranges, the word "estimate", a published backtest and a dated source on every card.
- **Noisy trend signals:** job-ad data is tech-heavy. Mitigation: capped adjustments, minimum counts, shrinkage towards no change, and a person approves each update.
- **Bias:** no demographic inputs; CI checks that eight names from Malaysia's communities get identical results.
- **Provider dependence and rate limits:** arXiv falls back to OpenAlex; RunPod falls back to

Gemini; everything runs without any AI key.

- **Free hosting:** the API can sleep when idle (about a minute to wake); the app shows a notice and retries.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: Tan Hong Sheng (Team Leader)

Date: 24 October 2026

Appendix: System Block Diagram

